```
C:\Users\Jane Eyre\Desktop\Python-Study>python
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> states = ['Alabama','Alaska','Arizona','Arkansas','California','Colorado',
...           'Connecticut','Delaware','Florida','Georgia','Hawaii','Idaho',
...           'Illinois','Indiana','Iowa','Kansas','Kentucky','Louisiana',
...           'Maine','Maryland','Massachusetts','Michigan','Minnesota',
...           'Mississippi','Missouri','Montana','Nebraska','Nevada',
...           'New Hampshire','New Jersey','New Mexico','New York',
...           'North Carolina','North Dakota','Ohio','Oklahoma','Oregon',
...           'Pennsylvania','Rhode Island','South Carolina','South Dakota',
...           'Tennessee','Texas','Utah','Vermont','Virginia','Washington',
...           'West Virginia','Wisconsin','Wyoming']
>>> states[0]
'Alabama'
>>> states[-1]
'Wyoming'
>>> for s in states:
...     if ' ' in s:
...         print(s)
...
New Hampshire
New Jersey
New Mexico
New York
North Carolina
North Dakota
Rhode Island
South Carolina
South Dakota
West Virginia
>>> [s for s in states if ' ' in s]
['New Hampshire', 'New Jersey', 'New Mexico', 'New York', 'North Carolina', 'North Dakota', 'Rhode Island',
'South Carolina','South Dakota', 'West Virginia']
>>> [s for s in states]
['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California', 'Colorado', 'Connecticut', 'Delaware', 'Florida',
'Georgia', 'Hawaii', 'Idaho', 'Illinois', 'Indiana', 'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine',
'Maryland', 'Massachusetts', 'Michigan', 'Minnesota', 'Mississippi', 'Missouri', 'Montana', 'Nebraska',
'Nevada', 'New Hampshire', 'New Jersey', 'New Mexico', 'NewYork', 'North Carolina', 'North Dakota', 'Ohio',
'Oklahoma', 'Oregon', 'Pennsylvania', 'Rhode Island', 'South Carolina', 'South Dakota', 'Tennessee', 'Texas',
'Utah', 'Vermont', 'Virginia', 'Washington', 'West Virginia', 'Wisconsin', 'Wyoming']
>>> [s.upper() for s in states]
['ALABAMA', 'ALASKA', 'ARIZONA', 'ARKANSAS', 'CALIFORNIA', 'COLORADO', 'CONNECTICUT', 'DELAWARE', 'FLORIDA',
'GEORGIA', 'HAWAII', 'IDAHO', 'ILLINOIS', 'INDIANA', 'IOWA', 'KANSAS', 'KENTUCKY', 'LOUISIANA', 'MAINE',
'MARYLAND', 'MASSACHUSETTS', 'MICHIGAN', 'MINNESOTA', 'MISSISSIPPI', 'MISSOURI', 'MONTANA', 'NEBRASKA',
'NEVADA', 'NEW HAMPSHIRE', 'NEW JERSEY', 'NEW MEXICO', 'NEWYORK', 'NORTH CAROLINA', 'NORTH DAKOTA', 'OHIO',
'OKLAHOMA', 'OREGON', 'PENNSYLVANIA', 'RHODE ISLAND', 'SOUTH CAROLINA', 'SOUTH DAKOTA', 'TENNESSEE', 'TEXAS',
'UTAH', 'VERMONT', 'VIRGINIA', 'WASHINGTON', 'WEST VIRGINIA', 'WISCONSIN', 'WYOMING']
>>> [s.replace(' ', '-') for s in states if ' ' in s]
['New-Hampshire', 'New-Jersey', 'New-Mexico', 'New-York', 'North-Carolina', 'North-Dakota', 'Rhode-Island',
'South-Carolina','South-Dakota', 'West-Virginia']
>>> [s for s in states if 'a' not in s.lower()]
['Connecticut', 'Illinois', 'Kentucky', 'Mississippi', 'Missouri', 'New Jersey', 'New Mexico', 'New York',
'Ohio', 'Oregon', 'Tennessee', 'Vermont', 'Wisconsin', 'Wyoming']
>>> [len(s) for s in states]
[7, 6, 7, 8, 10, 8, 11, 8, 7, 7, 6, 5, 8, 7, 4, 6, 8, 9, 5, 8, 13, 8, 9, 11, 8, 7, 8, 6, 13, 10, 10, 8, 14, 12,
4, 8, 6, 12, 12, 14, 12, 9, 5, 4, 7, 8, 10, 13, 9, 7]
>>> max([len(s) for s in states])
14
>>> [(len(s), s) for s in states]
[(7, 'Alabama'), (6, 'Alaska'), (7, 'Arizona'), (8, 'Arkansas'), (10, 'California'), (8, 'Colorado'), (11,
'Connecticut'), (8, 'Delaware'), (7, 'Florida'), (7, 'Georgia'), (6, 'Hawaii'), (5, 'Idaho'), (8, 'Illinois'),
(7, 'Indiana'), (4, 'Iowa'), (6, 'Kansas'), (8, 'Kentucky'), (9, 'Louisiana'), (5, 'Maine'), (8, 'Maryland'),
(13, 'Massachusetts'), (8, 'Michigan'), (9, 'Minnesota'), (11, 'Mississippi'), (8, 'Missouri'), (7, 'Montana'),
(8, 'Nebraska'), (6, 'Nevada'), (13, 'New Hampshire'), (10, 'New Jersey'), (10, 'New Mexico'), (8, 'New York'),
(14, 'North Carolina'), (12, 'North Dakota'), (4, 'Ohio'), (8, 'Oklahoma'), (6, 'Oregon'), (12, 'Pennsylvania'),
(12, 'Rhode Island'), (14, 'South Carolina'), (12, 'South Dakota'), (9, 'Tennessee'), (5, 'Texas'), (4, 'Utah'),
(7, 'Vermont'), (8, 'Virginia'), (10, 'Washington'), (13, 'West Virginia'), (9, 'Wisconsin'), (7, 'Wyoming')]
>>> sorted([(len(s), s) for s in states])
[(4, 'Iowa'), (4, 'Ohio'), (4, 'Utah'), (5, 'Idaho'), (5, 'Maine'), (5, 'Texas'), (6, 'Alaska'), (6, 'Hawaii'),
(6, 'Kansas'), (6, 'Nevada'), (6, 'Oregon'), (7, 'Alabama'), (7, 'Arizona'), (7, 'Florida'), (7, 'Georgia'), (7,
```

```
'Indiana'), (7, 'Montana'), (7, 'Vermont'), (7, 'Wyoming'), (8, 'Arkansas'), (8, 'Colorado'), (8, 'Delaware'), (8, 'Illinois'), (8, 'Kentucky'), (8, 'Maryland'), (8, 'Michigan'), (8, 'Missouri'), (8, 'Nebraska'), (8, 'New York'), (8, 'Oklahoma'), (8, 'Virginia'), (9, 'Louisiana'), (9, 'Minnesota'), (9, 'Tennessee'), (9, 'Wisconsin'), (10, 'California'), (10, 'New Jersey'), (10, 'New Mexico'), (10, 'Washington'), (11, 'Connecticut'), (11, 'Mississippi'), (12, 'North Dakota'), (12, 'Pennsylvania'), (12, 'Rhode Island'), (12, 'South Dakota'), (13, 'Massachusetts'), (13, 'New Hampshire'), (13, 'West Virginia'), (14, 'North Carolina'), (14, 'South Carolina')]
>>> sorted([2, 5, 22, 8, 145, 9])
[2, 5, 8, 9, 22, 145]
>>> sorted([2, 5, 22, 8, 145, 9], reverse=True)
[145, 22, 9, 8, 5, 2]
>>> sorted([(len(s), s) for s in states], reverse=True)
[(14, 'South Carolina'), (14, 'North Carolina'), (13, 'West Virginia'), (13, 'New Hampshire'), (13, 'Massachusetts'), (12, 'South Dakota'), (12, 'Rhode Island'), (12, 'Pennsylvania'), (12, 'North Dakota'), (11, 'Mississippi'), (11, 'Connecticut'), (10, 'Washington'), (10, 'New Mexico'), (10, 'New Jersey'), (10, 'California'), (9, 'Wisconsin'), (9, 'Tennessee'), (9, 'Minnesota'), (9, 'Louisiana'), (8, 'Virginia'), (8, 'Oklahoma'), (8, 'New York'), (8, 'Nebraska'), (8, 'Missouri'), (8, 'Michigan'), (8, 'Maryland'), (8, 'Kentucky'), (8, 'Illinois'), (8, 'Delaware'), (8, 'Colorado'), (8, 'Arkansas'), (7, 'Wyoming'), (7, 'Vermont'), (7, 'Montana'), (7, 'Indiana'), (7, 'Georgia'), (7, 'Florida'), (7, 'Arizona'), (7, 'Alabama'), (6, 'Oregon'), (6, 'Nevada'), (6, 'Kansas'), (6, 'Hawaii'), (6, 'Alaska'), (5, 'Texas'), (5, 'Maine'), (5, 'Idaho'), (4, 'Utah'), (4, 'Ohio'), (4, 'Iowa')]
>>> sorted([(len(s), s) for s in states], reverse=True)[:5]
[(14, 'South Carolina'), (14, 'North Carolina'), (13, 'West Virginia'), (13, 'New Hampshire'), (13, 'Massachusetts')]
```

```
>>> capital = {'al':'Montgomery', 'ak':'Juneau', 'az':'Phoenix',
...            'ar':'Little Rock', 'ca':'Sacramento', 'co':'Denver',
...            'ct':'Hartford', 'de':'Dover', 'fl':'Tallahassee', 'ga':'Atlanta',
...            'hi':'Honolulu', 'id':'Boise', 'il':'Springfield',
...            'in':'Indianapolis', 'ia':'Des Moines', 'ks':'Topeka',
...            'ky':'Frankfort', 'la':'Baton Rouge', 'me':'Augusta',
...            'md':'Annapolis', 'ma':'Boston', 'mi':'Lansing', 'mn':'St. Paul',
...            'ms':'Jackson', 'mo':'Jefferson City', 'mt':'Helena',
...            'ne':'Lincoln', 'nv':'Carson City', 'nh':'Concord', 'nj':'Trenton',
...            'nm':'Santa Fe', 'ny':'Albany', 'nc':'Raleigh', 'nd':'Bismarck',
...            'oh':'Columbus', 'ok':'Oklahoma City', 'or':'Salem',
...            'pa':'Harrisburg', 'ri':'Providence', 'sc':'Columbia',
...            'sd':'Pierre', 'tn':'Nashville', 'tx':'Austin',
...            'ut':'Salt Lake City', 'vt':'Montpelier', 'va':'Richmond',
...            'wa':'Olympia', 'wv':'Charleston', 'wi':'Madison', 'wy':'Cheyenne'}
>>>
>>>
>>> capital['pa']
'Harrisburg'
>>> capital.values()
dict_values(['Montgomery', 'Juneau', 'Phoenix', 'Little Rock', 'Sacramento', 'Denver', 'Hartford', 'Dover', 'Tallahassee', 'Atlanta', 'Honolulu', 'Boise', 'Springfield', 'Indianapolis', 'Des Moines', 'Topeka', 'Frankfort', 'Baton Rouge', 'Augusta', 'Annapolis', 'Boston', 'Lansing', 'St. Paul', 'Jackson', 'Jefferson City', 'Helena', 'Lincoln', 'Carson City', 'Concord', 'Trenton', 'Santa Fe', 'Albany', 'Raleigh', 'Bismarck', 'Columbus', 'Oklahoma City', 'Salem', 'Harrisburg', 'Providence', 'Columbia', 'Pierre', 'Nashville', 'Austin', 'Salt Lake City', 'Montpelier', 'Richmond', 'Olympia', 'Charleston', 'Madison', 'Cheyenne'])
>>>
>>> capital.keys()
dict_keys(['al', 'ak', 'az', 'ar', 'ca', 'co', 'ct', 'de', 'fl', 'ga', 'hi', 'id', 'il', 'in', 'ia', 'ks', 'ky', 'la', 'me', 'md', 'ma', 'mi', 'mn', 'ms', 'mo', 'mt', 'ne', 'nv', 'nh', 'nj', 'nm', 'ny', 'nc', 'nd', 'oh', 'ok', 'or', 'pa', 'ri', 'sc', 'sd', 'tn', 'tx', 'ut', 'vt', 'va', 'wa', 'wv', 'wi', 'wy'])
>>> capital.values()
dict_values(['Montgomery', 'Juneau', 'Phoenix', 'Little Rock', 'Sacramento', 'Denver', 'Hartford', 'Dover', 'Tallahassee', 'Atlanta', 'Honolulu', 'Boise', 'Springfield', 'Indianapolis', 'Des Moines', 'Topeka', 'Frankfort', 'Baton Rouge', 'Augusta', 'Annapolis', 'Boston', 'Lansing', 'St. Paul', 'Jackson', 'Jefferson City', 'Helena', 'Lincoln', 'Carson City', 'Concord', 'Trenton', 'Santa Fe', 'Albany', 'Raleigh', 'Bismarck', 'Columbus', 'Oklahoma City', 'Salem', 'Harrisburg', 'Providence', 'Columbia', 'Pierre', 'Nashville', 'Austin', 'Salt Lake City', 'Montpelier', 'Richmond', 'Olympia', 'Charleston', 'Madison', 'Cheyenne'])
>>> sorted(capital.values())
['Albany', 'Annapolis', 'Atlanta', 'Augusta', 'Austin', 'Baton Rouge', 'Bismarck', 'Boise', 'Boston', 'Carson City', 'Charleston', 'Cheyenne', 'Columbia', 'Columbus', 'Concord', 'Denver', 'Des Moines', 'Dover', 'Frankfort', 'Harrisburg', 'Hartford', 'Helena', 'Honolulu', 'Indianapolis', 'Jackson', 'Jefferson City', 'Juneau', 'Lansing', 'Lincoln', 'Little Rock', 'Madison', 'Montgomery', 'Montpelier', 'Nashville', 'Oklahoma City', 'Olympia', 'Phoenix', 'Pierre', 'Providence', 'Raleigh', 'Richmond', 'Sacramento', 'Salem', 'Salt Lake City', 'Santa Fe', 'Springfield', 'St. Paul', 'Tallahassee', 'Topeka', 'Trenton']
```

```
>>> sorted(capital.values(), reverse=True)
['Trenton', 'Topeka', 'Tallahassee', 'St. Paul', 'Springfield', 'Santa Fe', 'Salt Lake City', 'Salem',
'Sacramento', 'Richmond', 'Raleigh', 'Providence', 'Pierre', 'Phoenix', 'Olympia', 'Oklahoma City', 'Nashville',
'Montpelier', 'Montgomery', 'Madison', 'Little Rock', 'Lincoln', 'Lansing', 'Juneau', 'Jefferson City',
'Jackson', 'Indianapolis', 'Honolulu', 'Helena', 'Hartford', 'Harrisburg', 'Frankfort', 'Dover', 'Des Moines',
'Denver', 'Concord', 'Columbus', 'Columbia', 'Cheyenne', 'Charleston', 'Carson City', 'Boston', 'Boise',
'Bismarck', 'Baton Rouge', 'Austin', 'Augusta', 'Atlanta', 'Annapolis', 'Albany']
>>> capital['ma']
'Boston'
>>> capital.items()
dict_items([('al', 'Montgomery'), ('ak', 'Juneau'), ('az', 'Phoenix'), ('ar', 'Little Rock'), ('ca',
'Sacramento'), ('co', 'Denver'), ('ct', 'Hartford'), ('de', 'Dover'), ('fl', 'Tallahassee'), ('ga', 'Atlanta'),
('hi', 'Honolulu'), ('id', 'Boise'), ('il', 'Springfield'), ('in', 'Indianapolis'), ('ia', 'Des Moines'), ('ks',
'Topeka'), ('ky', 'Frankfort'), ('la', 'Baton Rouge'), ('me', 'Augusta'), ('md', 'Annapolis'), ('ma', 'Boston'),
('mi', 'Lansing'), ('mn', 'St. Paul'), ('ms', 'Jackson'), ('mo','Jefferson City'), ('mt', 'Helena'), ('ne',
'Lincoln'), ('nv', 'Carson City'), ('nh', 'Concord'), ('nj', 'Trenton'), ('nm', 'Santa Fe'), ('ny', 'Albany'),
('nc', 'Raleigh'), ('nd', 'Bismarck'), ('oh', 'Columbus'), ('ok', 'Oklahoma City'), ('or', 'Salem'), ('pa',
'Harrisburg'), ('ri', 'Providence'), ('sc', 'Columbia'), ('sd', 'Pierre'), ('tn', 'Nashville'), ('tx',
'Austin'), ('ut', 'Salt Lake City'), ('vt', 'Montpelier'), ('va', 'Richmond'), ('wa', 'Olympia'), ('wv',
'Charleston'), ('wi', 'Madison'), ('wy', 'Cheyenne')])
>>> [(s,c) for (s,c) in capital.items() if c=='Concord']
[('nh', 'Concord')]
>>> for x in capital:
...     print(x)
...
al
ak
az
ar
ca
co
ct
# ... results clipped! ...
wv
wi
wy
>>> [x for x in capital if 'k' in x]
['ak', 'ks', 'ky', 'ok']
>>> [x for x in capital.values() if 'k' in x]
['Little Rock', 'Topeka', 'Frankfort', 'Jackson', 'Bismarck', 'Oklahoma City', 'Salt Lake City']
>>> [x for x in capital.keys() if 'k' in x]
['ak', 'ks', 'ky', 'ok']
>>> [x for x in capital.items()]
[('al', 'Montgomery'), ('ak', 'Juneau'), ('az', 'Phoenix'), ('ar', 'Little Rock'), ('ca', 'Sacramento'), ('co',
'Denver'), ('ct', 'Hartford'), ('de', 'Dover'), ('fl', 'Tallahassee'), ('ga', 'Atlanta'), ('hi', 'Honolulu'),
('id', 'Boise'), ('il', 'Springfield'), ('in', 'Indianapolis'), ('ia', 'Des Moines'), ('ks', 'Topeka'), ('ky',
'Frankfort'), ('la', 'Baton Rouge'), ('me', 'Augusta'), ('md', 'Annapolis'), ('ma', 'Boston'), ('mi',
'Lansing'), ('mn', 'St. Paul'), ('ms', 'Jackson'), ('mo', 'JeffersonCity'), ('mt', 'Helena'), ('ne', 'Lincoln'),
('nv', 'Carson City'), ('nh', 'Concord'), ('nj', 'Trenton'), ('nm', 'Santa Fe'),('ny', 'Albany'), ('nc',
'Raleigh'), ('nd', 'Bismarck'), ('oh', 'Columbus'), ('ok', 'Oklahoma City'), ('or', 'Salem'),
('pa','Harrisburg'), ('ri', 'Providence'), ('sc', 'Columbia'), ('sd', 'Pierre'), ('tn', 'Nashville'), ('tx',
'Austin'), ('ut', 'Salt Lake City'), ('vt', 'Montpelier'), ('va', 'Richmond'), ('wa', 'Olympia'), ('wv',
'Charleston'), ('wi', 'Madison'), ('wy', 'Cheyenne')]
>>> capital.items()
dict_items([('al', 'Montgomery'), ('ak', 'Juneau'), ('az', 'Phoenix'), ('ar', 'Little Rock'), ('ca',
'Sacramento'), ('co', 'Denver'), ('ct', 'Hartford'), ('de', 'Dover'), ('fl', 'Tallahassee'), ('ga', 'Atlanta'),
('hi', 'Honolulu'), ('id', 'Boise'), ('il', 'Springfield'), ('in', 'Indianapolis'), ('ia', 'Des Moines'), ('ks',
'Topeka'), ('ky', 'Frankfort'), ('la', 'Baton Rouge'), ('me', 'Augusta'), ('md', 'Annapolis'), ('ma', 'Boston'),
('mi', 'Lansing'), ('mn', 'St. Paul'), ('ms', 'Jackson'), ('mo','Jefferson City'), ('mt', 'Helena'), ('ne',
'Lincoln'), ('nv', 'Carson City'), ('nh', 'Concord'), ('nj', 'Trenton'), ('nm', 'Santa Fe'), ('ny', 'Albany'),
('nc', 'Raleigh'), ('nd', 'Bismarck'), ('oh', 'Columbus'), ('ok', 'Oklahoma City'), ('or', 'Salem'), ('pa',
'Harrisburg'), ('ri', 'Providence'), ('sc', 'Columbia'), ('sd', 'Pierre'), ('tn', 'Nashville'), ('tx',
'Austin'), ('ut', 'Salt Lake City'), ('vt', 'Montpelier'), ('va', 'Richmond'), ('wa', 'Olympia'), ('wv',
'Charleston'), ('wi', 'Madison'), ('wy', 'Cheyenne')])
>>> [(s,c) for (s,c) in capital.items() if 'k' in s and 'k' in c]
[('ks', 'Topeka'), ('ky', 'Frankfort'), ('ok', 'Oklahoma City')]
```

```
>>> popul={'al':4833722, 'ak':735132, 'az':6626624, 'ar':2959373, 'ca':38332521,
...         'co':5268367, 'ct':3596080, 'de':925749, 'fl':19552860, 'ga':9992167,
```

```
...          'hi':1404054, 'id':1612136, 'il':12882135, 'in':6570902, 'ia':3090416,
...          'ks':2893957, 'ky':4395295, 'la':4625470, 'me':1328302, 'md':5928814,
...          'ma':6692824, 'mi':9895622, 'mn':5420380, 'ms':2991207, 'mo':6021988,
...          'mt':1015165, 'ne':1868516, 'nv':2790136, 'nh':1323459, 'nj':8899339,
...          'nm':2085287, 'ny':19651127, 'nc':9848060, 'nd':723393, 'oh':11570808,
...          'ok':3850568, 'or':3930065, 'pa':12773801, 'ri':1051511, 'sc':4774839,
...          'sd':844877, 'tn':6495978, 'tx':26448193, 'ut':2900872, 'vt':626630,
...          'va':8260405, 'wa':6971406, 'wv':1854304, 'wi':5742713, 'wy':582658}
>>>
>>> popul['pa']
12773801
>>> for s in popul:
...     if popul[s] > popul['pa']:
...         print(s, popul[s])
...
ca 38332521
fl 19552860
il 12882135
ny 19651127
tx 26448193
>>> [s for s in popul if popul[s] > popul['pa']]
['ca', 'fl', 'il', 'ny', 'tx']
>>> [(s, popul[s]) for s in popul if popul[s] > popul['pa']]
[('ca', 38332521), ('fl', 19552860), ('il', 12882135), ('ny', 19651127), ('tx', 26448193)]
>>> [(s,p) for (s,p) in popul.items() if p > popul['pa']]
[('ca', 38332521), ('fl', 19552860), ('il', 12882135), ('ny', 19651127), ('tx', 26448193)]
>>> [(s,p, p/popul['pa']) for (s,p) in popul.items() if p > popul['pa']]
[('ca', 38332521, 3.0008703752313033), ('fl', 19552860, 1.530700219926708), ('il', 12882135, 1.00848095253715),
('ny', 19651127, 1.5383930750134591), ('tx', 26448193, 2.0705029771483052)]
>>> sorted([(p/popul['pa'], s,p) for (s,p) in popul.items() if p > popul['pa']])
[(1.00848095253715, 'il', 12882135), (1.530700219926708, 'fl', 19552860), (1.5383930750134591, 'ny', 19651127),
(2.0705029771483052, 'tx', 26448193), (3.0008703752313033, 'ca', 38332521)]
>>> sorted([(p/popul['pa'], s,p) for (s,p) in popul.items() if p > popul['pa']], reverse=True)
[(3.0008703752313033, 'ca', 38332521), (2.0705029771483052, 'tx', 26448193), (1.5383930750134591, 'ny',
19651127), (1.530700219926708, 'fl', 19552860), (1.00848095253715, 'il', 12882135)]
>>> [(s,p) for (s,p) in popul.items() if p > popul['pa']]
[('ca', 38332521), ('fl', 19552860), ('il', 12882135), ('ny', 19651127), ('tx', 26448193)]
>>> dict([(s,p) for (s,p) in popul.items() if p > popul['pa']])
{'ca': 38332521, 'fl': 19552860, 'il': 12882135, 'ny': 19651127, 'tx': 26448193}


>>> footuple = [('a', 'apple'), ('b', 'banana'), ('c', 'carrot')]
>>> footuple
[('a', 'apple'), ('b', 'banana'), ('c', 'carrot')]
>>> [y for (x,y) in footuple]
['a', 'b', 'c']
>>> [(y,x) for (x,y) in footuple]
[('apple', 'a'), ('banana', 'b'), ('carrot', 'c')]
>>> dict(footuple)
{'a': 'apple', 'b': 'banana', 'c': 'carrot'}
>>> foodict = dict(footuple)
>>> foodict
{'a': 'apple', 'b': 'banana', 'c': 'carrot'}
>>> foodict['a']
'apple'


# Copy over gettyburg address string
>>> getty
'Four score and seven years ago our fathers brought forth on this continent a new nation, conceived in liberty,
and dedicatedto the proposition that all men are created equal.\n\nNow we are engaged in a great civil war,
testing whether that nation, or any nation, so conceived and so dedicated, can long endure. We are met on a
great battle-field of that war. We have come to dedicate a portion of that field, as a final resting place for
those who here gave their lives that that nation might live. Itis altogether fitting and proper that we should
do this.\n\nBut, in a larger sense, we can not dedicate, we can not consecrate, we can not hallow this ground.
The brave men, living and dead, who struggled here, have consecrated it, far above our poor power to add or
detract. The world will little note, nor long remember what we say here, but it can never forget what they
didhere. It is for us the living, rather, to be dedicated here to the unfinished work which they who fought here
have thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining before us
- that from these honored dead we take increased devotion to that cause for which they gave the last full
measure of devotion - that we here highly resolvethat these dead shall not have died in vain - that this nation,
under God, shall have a new birth of freedom - and that government of the people, by the people, for the people,
```

```
shall not perish from the earth.'
>>> getty_toks = getty.split()
>>> getty_toks[:30]
['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation,', 'conceived', 'in', 'liberty,', 'and', 'dedicated', 'to', 'the',
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal.']
>>> len(getty_toks)
275
>>> [t for t in getty_toks if len(t) <=3]
['and', 'ago', 'our', 'on', 'a', 'new', 'in', 'and', 'to', 'the', 'all', 'men', 'are', 'Now', 'we', 'are', 'in',
'a', 'or', 'any', 'so', 'and', 'so', 'can', 'We', 'are', 'met', 'on', 'a', 'of', 'We', 'to', 'a', 'of', 'as',
'a', 'for', 'who', 'It', 'is', 'and', 'we', 'do', 'in', 'a', 'we', 'can', 'not', 'we', 'can', 'not', 'we',
'can', 'not', 'The', 'and', 'who', 'it,', 'far', 'our', 'to', 'add', 'or', 'The', 'nor', 'we', 'say', 'but',
'it', 'can', 'did', 'It', 'is', 'for', 'us', 'the', 'to', 'be', 'to', 'the', 'who', 'far', 'so', 'It', 'is',
'for', 'us', 'to', 'be', 'to', 'the', 'us', '-', 'we', 'to', 'for', 'the', 'of', '-', 'we', 'not', 'in', '-',
'a', 'new', 'of', '-', 'and', 'of', 'the', 'by', 'the', 'for', 'the', 'not', 'the']
>>> [t for t in getty_toks if len(t) > 5]
['fathers', 'brought', 'continent', 'nation,', 'conceived', 'liberty,', 'dedicated', 'proposition', 'created',
'equal.', 'engaged', 'testing', 'whether', 'nation,', 'nation,', 'conceived', 'dedicated,', 'endure.', 'battle-
field', 'dedicate', 'portion', 'field,', 'resting', 'nation', 'altogether', 'fitting', 'proper', 'should',
'larger', 'sense,', 'dedicate,', 'consecrate,', 'hallow', 'ground.', 'living', 'struggled', 'consecrated',
'detract.', 'little', 'remember', 'forget', 'living,', 'rather,', 'dedicated', 'unfinished', 'fought',
'advanced.', 'rather', 'dedicated', 'remaining', 'before', 'honored', 'increased', 'devotion', 'measure',
'devotion', 'highly', 'resolve', 'nation,', 'freedom', 'government', 'people,', 'people,', 'people,', 'perish',
'earth.']
>>> [(len(t), t) for t in getty_toks]
[(4, 'Four'), (5, 'score'), (3, 'and'), (5, 'seven'), (5, 'years'), (3, 'ago'), (3, 'our'), (7, 'fathers'), (7,
'brought'), (5, 'forth'), (2, 'on'), (4, 'this'), (9, 'continent'), (1, 'a'), (3, 'new'), (7, 'nation,'), (9,
'conceived'), (2, 'in'), (8, 'liberty,'), (3, 'and'), (9, 'dedicated'), (2, 'to'), (3, 'the'), (11,
'proposition'), (4, 'that'), (3, 'all'), (3, 'men'), (3, 'are'), (7, 'created'), (6, 'equal.'), (3, 'Now'), (2,
'we'), (3, 'are'), (7, 'engaged'), (2, 'in'), (1, 'a'), (5, 'great'),(5, 'civil'), (4, 'war,'), (7, 'testing'),
(7, 'whether'), (4, 'that'), (7, 'nation,'), (2, 'or'), (3, 'any'), (7, 'nation,'), (2, 'so'), (9, 'conceived'),
(3, 'and'), (2, 'so'), (10, 'dedicated,'), (3, 'can'), (4, 'long'), (7, 'endure.'), (2, 'We'), (3, 'are'), (3,
'met'), (2, 'on'), (1, 'a'), (5, 'great'), (12, 'battle-field'), (2, 'of'), (4, 'that'), (4, 'war.'), (2, 'We'),
(4, 'have'), (4, 'come'), (2, 'to'), (8, 'dedicate'), (1, 'a'), (7, 'portion'), (2, 'of'), (4, 'that'), (6,
'field,'), (2, 'as'), (1, 'a'), (5, 'final'), (7, 'resting'), (5, 'place'), (3, 'for'), (5, 'those'), (3,
'who'), (4, 'here'), (4, 'gave'), (5, 'their'), (5, 'lives'), (4, 'that'), (4, 'that'), (6, 'nation'), (5,
'might'), (5, 'live.'), (2, 'It'), (2, 'is'), (10, 'altogether'), (7, 'fitting'), (3, 'and'), (6, 'proper'), (4,
'that'), (2, 'we'), (6, 'should'), (2, 'do'), (5, 'this.'), (4, 'But,'), (2, 'in'), (1, 'a'), (6, 'larger'), (6,
'sense,'), (2, 'we'), (3, 'can'), (3, 'not'), (9, 'dedicate,'), (2, 'we'), (3, 'can'), (3, 'not'), (11,
'consecrate,'), (2, 'we'), (3, 'can'), (3, 'not'), (6, 'hallow'), (4, 'this'), (7, 'ground.'), (3, 'The'), (5,
'brave'), (4, 'men,'), (6, 'living'), (3, 'and'), (5, 'dead,'), (3, 'who'), (9, 'struggled'), (5, 'here,'), (4,
'have'), (11, 'consecrated'), (3, 'it,'), (3, 'far'), (5, 'above'), (3, 'our'), (4, 'poor'), (5, 'power'), (2,
'to'), (3, 'add'), (2, 'or'), (8, 'detract.'), (3, 'The'), (5, 'world'), (4, 'will'), (6, 'little'), (5,
'note,'), (3, 'nor'), (4, 'long'), (8, 'remember'), (4, 'what'), (2, 'we'), (3, 'say'), (5, 'here,'), (3,
'but'), (2, 'it'), (3, 'can'), (5, 'never'), (6, 'forget'), (4, 'what'), (4, 'they'), (3, 'did'), (5, 'here.'),
(2, 'It'), (2, 'is'), (3, 'for'), (2, 'us'), (3, 'the'), (7, 'living,'), (7, 'rather,'), (2, 'to'), (2, 'be'),
(9, 'dedicated'), (4, 'here'), (2, 'to'), (3, 'the'), (10, 'unfinished'), (4, 'work'), (5,'which'), (4, 'they'),
(3, 'who'), (6, 'fought'), (4, 'here'), (4, 'have'), (4, 'thus'), (3, 'far'), (2, 'so'), (5, 'nobly'),(9,
'advanced.'), (2, 'It'), (2, 'is'), (6, 'rather'), (3, 'for'), (2, 'us'), (2, 'to'), (2, 'be'), (4, 'here'), (9,
'dedicated'), (2, 'to'), (3, 'the'), (5, 'great'), (4, 'task'), (9, 'remaining'), (6, 'before'), (2, 'us'), (1,
'-'), (4, 'that'), (4,'from'), (5, 'these'), (7, 'honored'), (4, 'dead'), (2, 'we'), (4, 'take'), (9,
'increased'), (8, 'devotion'), (2, 'to'), (4,'that'), (5, 'cause'), (3, 'for'), (5, 'which'), (4, 'they'), (4,
'gave'), (3, 'the'), (4, 'last'), (4, 'full'), (7, 'measure'), (2, 'of'), (8, 'devotion'), (1, '-'), (4,
'that'), (2, 'we'), (4, 'here'), (6, 'highly'), (7, 'resolve'), (4, 'that'), (5,'these'), (4, 'dead'), (5,
'shall'), (3, 'not'), (4, 'have'), (4, 'died'), (2, 'in'), (4, 'vain'), (1, '-'), (4, 'that'), (4,'this'), (7,
'nation,'), (5, 'under'), (4, 'God,'), (5, 'shall'), (4, 'have'), (1, 'a'), (3, 'new'), (5, 'birth'), (2, 'of'),
(7, 'freedom'), (1, '-'), (3, 'and'), (4, 'that'), (10, 'government'), (2, 'of'), (3, 'the'), (7, 'people,'),
(2, 'by'), (3, 'the'), (7, 'people,'), (3, 'for'), (3, 'the'), (7, 'people,'), (5, 'shall'), (3, 'not'), (6,
'perish'), (4, 'from'), (3, 'the'), (6, 'earth.')]
>>> sorted([(len(t), t) for t in getty_toks], reverse=True)[:10]
[(12, 'battle-field'), (11, 'proposition'), (11, 'consecrated'), (11, 'consecrate,'), (10, 'unfinished'), (10,
'government'),(10, 'dedicated,'), (10, 'altogether'), (9, 'struggled'), (9, 'remaining')]
>>> [t for t in getty_toks if not t.isalnum()]
['nation,', 'liberty,', 'equal.', 'war,', 'nation,', 'nation,', 'dedicated,', 'endure.', 'battle-field', 'war.',
'field,', 'live.', 'this.', 'But,', 'sense,', 'dedicate,', 'consecrate,', 'ground.', 'men,', 'dead,', 'here,',
'it,', 'detract.', 'note,','here,', 'here.', 'living,', 'rather,', 'advanced.', '-', '-', '-', 'nation,',
'God,', '-', 'people,', 'people,', 'people,', 'earth.']
>>> [t for t in getty_toks if t[0].isupper()]
['Four', 'Now', 'We', 'We', 'It', 'But,', 'The', 'The', 'It', 'It', 'God,']
>>> [t for t in getty_toks if len(t) > 10 and '-' not in t]
['proposition', 'consecrate,', 'consecrated']
```

```
>>> [t for t in getty_toks if len(t) > 10 and '-' in t]
['battle-field']
>>> sent = 'We love school.'
>>> sent
'We love school.'
>>> sent.split(' ')
['We', 'love', 'school.']
>>> sent.split()
['We', 'love', 'school.']
>>> sent.replace('.', '')
'We love school'
>>> sent.replace('.', '').split()
['We', 'love', 'school']
>>> sent = 'We love school, and we love our teacher.'
>>> sent.replace('.', '').split()
['We', 'love', 'school,', 'and', 'we', 'love', 'our', 'teacher']
>>> sent.replace('.', '').replace(',', '').split()
['We', 'love', 'school', 'and', 'we', 'love', 'our', 'teacher']
>>> getty_toks
['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation,', 'conceived', 'in', 'liberty,', 'and', 'dedicated', 'to', 'the',
'proposition', 'that', 'all', 'men', 'are', 'created', 'equal.', 'Now', 'we', 'are', 'engaged', 'in', 'a',
'great', 'civil', 'war,', 'testing', 'whether', 'that', 'nation,', 'or', 'any', 'nation,', 'so', 'conceived',
'and', 'so', 'dedicated,', 'can', 'long', 'endure.', 'We', 'are', 'met', 'on', 'a', 'great', 'battle-field',
'of', 'that', 'war.', 'We', 'have', 'come', 'to', 'dedicate', 'a', 'portion', 'of', 'that', 'field,','as', 'a',
'final', 'resting', 'place', 'for', 'those', 'who', 'here', 'gave', 'their', 'lives', 'that', 'that', 'nation',
'might', 'live.', 'It', 'is', 'altogether', 'fitting', 'and', 'proper', 'that', 'we', 'should', 'do', 'this.',
'But,', 'in', 'a', 'larger', 'sense,', 'we', 'can', 'not', 'dedicate,', 'we', 'can', 'not', 'consecrate,', 'we',
'can', 'not', 'hallow', 'this', 'ground.', 'The', 'brave', 'men,', 'living', 'and', 'dead,', 'who', 'struggled',
'here,', 'have', 'consecrated', 'it,', 'far', 'above', 'our', 'poor', 'power', 'to', 'add', 'or', 'detract.',
'The', 'world', 'will', 'little', 'note,', 'nor', 'long', 'remember', 'what', 'we', 'say', 'here,', 'but', 'it',
'can', 'never', 'forget', 'what', 'they', 'did', 'here.', 'It', 'is', 'for', 'us', 'the', 'living,', 'rather,',
'to', 'be', 'dedicated', 'here', 'to', 'the', 'unfinished', 'work', 'which', 'they', 'who', 'fought', 'here',
'have', 'thus', 'far', 'so', 'nobly', 'advanced.', 'It', 'is', 'rather', 'for', 'us', 'to', 'be', 'here',
'dedicated', 'to', 'the', 'great', 'task', 'remaining', 'before', 'us', '-', 'that', 'from', 'these', 'honored',
'dead', 'we', 'take', 'increased', 'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the',
'last', 'full', 'measure','of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that', 'these',
'dead', 'shall', 'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation,', 'under', 'God,', 'shall',
'have', 'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that', 'government', 'of', 'the', 'people,', 'by',
'the', 'people,', 'for', 'the', 'people,', 'shall', 'not', 'perish', 'from', 'the', 'earth.']
>>> getty_toks.replace('.', '')       # can't use .replace() on a list
Traceback (most recent call last):
  File "", line 1, in
    getty_toks.replace('.', '')
    ^^^^^^^^^^^^^^^^^^
AttributeError: 'list' object has no attribute 'replace'
>>> [t.upper() for t in getty_toks]   # use list comprehension
['FOUR', 'SCORE', 'AND', 'SEVEN', 'YEARS', 'AGO', 'OUR', 'FATHERS', 'BROUGHT', 'FORTH', 'ON', 'THIS',
'CONTINENT', 'A', 'NEW', 'NATION,', 'CONCEIVED', 'IN', 'LIBERTY,', 'AND', 'DEDICATED', 'TO', 'THE',
'PROPOSITION', 'THAT', 'ALL', 'MEN', 'ARE', 'CREATED', 'EQUAL.', 'NOW', 'WE', 'ARE', 'ENGAGED', 'IN', 'A',
'GREAT', 'CIVIL', 'WAR,', 'TESTING', 'WHETHER', 'THAT', 'NATION,', 'OR', 'ANY', 'NATION,', 'SO', 'CONCEIVED',
'AND', 'SO', 'DEDICATED,', 'CAN', 'LONG', 'ENDURE.', 'WE', 'ARE', 'MET', 'ON', 'A', 'GREAT', 'BATTLE-FIELD',
'OF', 'THAT', 'WAR.', 'WE', 'HAVE', 'COME', 'TO', 'DEDICATE', 'A', 'PORTION', 'OF', 'THAT', 'FIELD,','AS', 'A',
'FINAL', 'RESTING', 'PLACE', 'FOR', 'THOSE', 'WHO', 'HERE', 'GAVE', 'THEIR', 'LIVES', 'THAT', 'THAT', 'NATION',
'MIGHT', 'LIVE.', 'IT', 'IS', 'ALTOGETHER', 'FITTING', 'AND', 'PROPER', 'THAT', 'WE', 'SHOULD', 'DO', 'THIS.',
'BUT,', 'IN', 'A', 'LARGER', 'SENSE,', 'WE', 'CAN', 'NOT', 'DEDICATE,', 'WE', 'CAN', 'NOT', 'CONSECRATE,', 'WE',
'CAN', 'NOT', 'HALLOW', 'THIS', 'GROUND.', 'THE', 'BRAVE', 'MEN,', 'LIVING', 'AND', 'DEAD,', 'WHO', 'STRUGGLED',
'HERE,', 'HAVE', 'CONSECRATED', 'IT,', 'FAR', 'ABOVE', 'OUR', 'POOR', 'POWER', 'TO', 'ADD', 'OR', 'DETRACT.',
'THE', 'WORLD', 'WILL', 'LITTLE', 'NOTE,', 'NOR', 'LONG', 'REMEMBER', 'WHAT', 'WE', 'SAY', 'HERE,', 'BUT', 'IT',
'CAN', 'NEVER', 'FORGET', 'WHAT', 'THEY', 'DID', 'HERE.', 'IT', 'IS', 'FOR', 'US', 'THE', 'LIVING,', 'RATHER,',
'TO', 'BE', 'DEDICATED', 'HERE', 'TO', 'THE', 'UNFINISHED', 'WORK', 'WHICH', 'THEY', 'WHO', 'FOUGHT', 'HERE',
'HAVE', 'THUS', 'FAR', 'SO', 'NOBLY', 'ADVANCED.', 'IT', 'IS', 'RATHER', 'FOR', 'US', 'TO', 'BE', 'HERE',
'DEDICATED', 'TO', 'THE', 'GREAT', 'TASK', 'REMAINING', 'BEFORE', 'US', '-', 'THAT', 'FROM', 'THESE', 'HONORED',
'DEAD', 'WE', 'TAKE', 'INCREASED', 'DEVOTION', 'TO', 'THAT', 'CAUSE', 'FOR', 'WHICH', 'THEY', 'GAVE', 'THE',
'LAST', 'FULL', 'MEASURE','OF', 'DEVOTION', '-', 'THAT', 'WE', 'HERE', 'HIGHLY', 'RESOLVE', 'THAT', 'THESE',
'DEAD', 'SHALL', 'NOT', 'HAVE', 'DIED', 'IN', 'VAIN', '-', 'THAT', 'THIS', 'NATION,', 'UNDER', 'GOD,', 'SHALL',
'HAVE', 'A', 'NEW', 'BIRTH', 'OF', 'FREEDOM', '-', 'AND', 'THAT', 'GOVERNMENT', 'OF', 'THE', 'PEOPLE,', 'BY',
'THE', 'PEOPLE,', 'FOR', 'THE', 'PEOPLE,', 'SHALL', 'NOT', 'PERISH', 'FROM', 'THE', 'EARTH.']
>>> [t.replace(',', '') for t in getty_toks]
['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the', 'proposition',
'that', 'all', 'men', 'are', 'created', 'equal.', 'Now', 'we', 'are', 'engaged', 'in', 'a', 'great', 'civil',
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'war', 'testing', 'whether', 'that', 'nation', 'or','any', 'nation', 'so', 'conceived', 'and', 'so',
'dedicated', 'can', 'long', 'endure.', 'We', 'are', 'met', 'on', 'a', 'great', 'battle-field', 'of', 'that',
'war.', 'We', 'have', 'come', 'to', 'dedicate', 'a', 'portion', 'of', 'that', 'field', 'as', 'a', 'final',
'resting', 'place', 'for', 'those', 'who', 'here', 'gave', 'their', 'lives', 'that', 'that', 'nation', 'might',
'live.', 'It', 'is', 'altogether', 'fitting', 'and', 'proper', 'that', 'we', 'should', 'do', 'this.', 'But',
'in', 'a', 'larger', 'sense', 'we', 'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate', 'we', 'can',
'not', 'hallow', 'this', 'ground.', 'The', 'brave', 'men', 'living', 'and', 'dead', 'who', 'struggled', 'here',
'have', 'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to', 'add', 'or', 'detract.', 'The',
'world', 'will', 'little', 'note', 'nor', 'long', 'remember', 'what', 'we', 'say', 'here', 'but', 'it', 'can',
'never', 'forget', 'what', 'they', 'did', 'here.', 'It', 'is', 'for', 'us', 'the','living', 'rather', 'to',
'be', 'dedicated', 'here', 'to', 'the', 'unfinished', 'work', 'which', 'they', 'who', 'fought', 'here', 'have',
'thus', 'far', 'so', 'nobly', 'advanced.', 'It', 'is', 'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated',
'to', 'the', 'great', 'task', 'remaining', 'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead',
'we', 'take', 'increased', 'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the', 'last',
'full', 'measure', 'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that', 'these', 'dead',
'shall', 'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation', 'under', 'God', 'shall', 'have',
'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that', 'government', 'of', 'the', 'people', 'by', 'the',
'people', 'for', 'the', 'people', 'shall', 'not', 'perish', 'from', 'the', 'earth.']
>>> [t.replace(',', '').replace('.', '') for t in getty_toks]
['Four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the', 'proposition',
'that', 'all', 'men', 'are', 'created', 'equal', 'Now', 'we', 'are', 'engaged', 'in', 'a', 'great', 'civil',
'war', 'testing', 'whether', 'that', 'nation', 'or', 'any', 'nation', 'so', 'conceived', 'and', 'so',
'dedicated', 'can', 'long', 'endure', 'We', 'are', 'met', 'on', 'a', 'great','battle-field', 'of', 'that',
'war', 'We', 'have', 'come', 'to', 'dedicate', 'a', 'portion', 'of', 'that', 'field', 'as', 'a', 'final',
'resting', 'place', 'for', 'those', 'who', 'here', 'gave', 'their', 'lives', 'that', 'that', 'nation', 'might',
'live', 'It', 'is', 'altogether', 'fitting', 'and', 'proper', 'that', 'we', 'should', 'do', 'this', 'But', 'in',
'a', 'larger', 'sense', 'we', 'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate', 'we', 'can', 'not',
'hallow', 'this', 'ground', 'The', 'brave', 'men', 'living', 'and', 'dead', 'who', 'struggled', 'here', 'have',
'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to', 'add', 'or', 'detract', 'The', 'world',
'will', 'little', 'note', 'nor', 'long', 'remember', 'what', 'we', 'say', 'here', 'but', 'it', 'can', 'never',
'forget', 'what', 'they', 'did', 'here', 'It', 'is', 'for', 'us', 'the', 'living', 'rather', 'to', 'be',
'dedicated', 'here', 'to', 'the', 'unfinished', 'work', 'which', 'they', 'who', 'fought', 'here', 'have',
'thus', 'far', 'so', 'nobly', 'advanced', 'It', 'is', 'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated',
'to', 'the', 'great', 'task', 'remaining', 'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead',
'we', 'take', 'increased', 'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the', 'last',
'full', 'measure', 'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that', 'these', 'dead',
'shall', 'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation', 'under', 'God', 'shall', 'have',
'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that', 'government', 'of', 'the', 'people', 'by', 'the',
'people', 'for', 'the', 'people', 'shall', 'not', 'perish', 'from', 'the', 'earth']
>>> [t.replace(',', '').replace('.', '').lower() for t in getty_toks]
['four', 'score', 'and', 'seven', 'years', 'ago', 'our', 'fathers', 'brought', 'forth', 'on', 'this',
'continent', 'a', 'new', 'nation', 'conceived', 'in', 'liberty', 'and', 'dedicated', 'to', 'the', 'proposition',
'that', 'all', 'men', 'are', 'created', 'equal', 'now', 'we', 'are', 'engaged', 'in', 'a', 'great', 'civil',
'war', 'testing', 'whether', 'that', 'nation', 'or', 'any', 'nation', 'so', 'conceived', 'and', 'so',
'dedicated', 'can', 'long', 'endure', 'we', 'are', 'met', 'on', 'a', 'great','battle-field', 'of', 'that',
'war', 'we', 'have', 'come', 'to', 'dedicate', 'a', 'portion', 'of', 'that', 'field', 'as', 'a', 'final',
'resting', 'place', 'for', 'those', 'who', 'here', 'gave', 'their', 'lives', 'that', 'that', 'nation', 'might',
'live', 'it', 'is', 'altogether', 'fitting', 'and', 'proper', 'that', 'we', 'should', 'do', 'this', 'but', 'in',
'a', 'larger', 'sense', 'we', 'can', 'not', 'dedicate', 'we', 'can', 'not', 'consecrate', 'we', 'can', 'not',
'hallow', 'this', 'ground', 'the', 'brave', 'men', 'living', 'and', 'dead', 'who', 'struggled', 'here', 'have',
'consecrated', 'it', 'far', 'above', 'our', 'poor', 'power', 'to', 'add', 'or', 'detract', 'the', 'world',
'will', 'little', 'note', 'nor', 'long', 'remember', 'what', 'we', 'say', 'here', 'but', 'it', 'can', 'never',
'forget', 'what', 'they', 'did', 'here', 'it', 'is', 'for', 'us', 'the', 'living', 'rather', 'to', 'be',
'dedicated', 'here', 'to', 'the', 'unfinished', 'work', 'which', 'they', 'who', 'fought', 'here', 'have',
'thus', 'far', 'so', 'nobly', 'advanced', 'it', 'is', 'rather', 'for', 'us', 'to', 'be', 'here', 'dedicated',
'to', 'the', 'great', 'task', 'remaining', 'before', 'us', '-', 'that', 'from', 'these', 'honored', 'dead',
'we', 'take', 'increased', 'devotion', 'to', 'that', 'cause', 'for', 'which', 'they', 'gave', 'the', 'last',
'full', 'measure', 'of', 'devotion', '-', 'that', 'we', 'here', 'highly', 'resolve', 'that', 'these', 'dead',
'shall', 'not', 'have', 'died', 'in', 'vain', '-', 'that', 'this', 'nation', 'under', 'god', 'shall', 'have',
'a', 'new', 'birth', 'of', 'freedom', '-', 'and', 'that', 'government', 'of', 'the', 'people', 'by', 'the',
'people', 'for', 'the', 'people', 'shall', 'not', 'perish', 'from', 'the', 'earth']
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